

Flight Experience Feedback Analysis

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Abstract—This project explores airline passenger satisfaction by analyzing customer reviews through sentiment analysis. Three different methods were tested: (1) word-score sentiment analysis using the AFINN lexicon, (2) rule-based sentiment analysis with SentiWordNet, and (3) a deep learning approach using the cardiffnlp/twitter-xlm-roberta-base-sentiment model. A hybrid labeling process was applied to 22,098 scraped reviews, combining results from TextBlob, VADER, and keyword matching, with a strict agreement rule to ensure consistency. Ambiguous reviews were pseudo-labeled using the distilbert-base-uncased-finetuned-sst-2-english model. To tackle the issue of having too few neutral reviews, GPT-4 was used to generate synthetic neutral samples, resulting in a validation set of 12,000 reviews. This final validation set included both hybrid-labeled and synthetic data, with a separate test set of 3,000 reviews drawn from unused samples. Evaluation metrics like accuracy, precision, recall, and F1-score were used to assess the models. The results showed that the word-score method achieved the highest test accuracy (80.8%) and F1-score (80.6%), slightly outperforming the deep learning model. The rule-based method consistently performed the worst. These findings suggest that, with well-constructed datasets, traditional lexicon-based methods can be competitive with more complex deep learning models.

I. INTRODUCTION

In recent years, structured frameworks for sentiment analysis have become powerful tools for evaluating airline service quality. These frameworks help airlines gain valuable insights into customer satisfaction, allowing them to refine their services and stay competitive in an increasingly crowded industry [1]. With the growing reliance on data-driven decisions, understanding passenger sentiment has become essential for improving service offerings and the overall customer experience.

Sentiment analysis, a key technique in natural language processing (NLP), is widely used to extract subjective information from text and classify sentiments as positive, negative, or neutral. This approach is particularly useful for analyzing airline passenger reviews, as it reveals important details about customer satisfaction and dissatisfaction. However, the challenge lies in selecting the most effective method to analyze the diverse nature of these reviews. This study evaluates three

different sentiment analysis techniques: (1) Word-Score Sentiment Scoring with the AFINN lexicon, (2) Rule-Based Sentiment Analysis using SentiWordNet, and (3) Deep Learning-Based Sentiment Analysis with the cardiffnlp/twitter-xlm-roberta-base-sentiment model. By comparing these methods, the study aims to identify the best approach for extracting actionable insights from passenger reviews.

As competition in the airline industry grows, understanding passenger expectations and perceptions is crucial for addressing service gaps and improving customer satisfaction. Recent research has shown the potential of unsupervised text analytics to leverage online customer reviews—an easily accessible and cost-effective resource—to capture customer opinions. Techniques like pLSA, LDA-VI, and LDA-GS, combined with sentiment analysis, have proven effective in providing airlines with insights to enhance service quality. However, previous studies often rely on individual techniques that may not fully capture the complexities of customer feedback. This study expands on these methods by integrating sentiment analysis with advanced topic modeling to offer a more comprehensive tool for understanding passenger feedback [2].

The growing use of social media in the airline industry for real-time customer feedback has opened up new possibilities for sentiment analysis. Recent studies have examined how User-Generated Content (UGC), such as Twitter data, can help identify public sentiment and pinpoint service disruptions. For example, by analyzing over 1.3 million Tweets from U.S. airline passengers, researchers demonstrated how sentiment and lexical analysis can uncover abnormal sentiment trends, offering actionable insights for improving customer relationship management. This study builds on that research by incorporating financial metrics like Bollinger Bands to better visualize and interpret sentiment trends, giving airlines a more detailed understanding of customer sentiment [3].

Moreover, the combination of machine learning and causal analysis has shown significant potential in enhancing customer satisfaction in the airline industry. Data-driven approaches can offer deeper insights into customer needs, helping airlines

align their services with passenger expectations. This study harnesses the power of machine learning and causal analysis to provide airlines with tools for making more informed, data-driven decisions that improve service quality and customer satisfaction [4].

Sentiment analysis (SA) has garnered significant attention due to its wide range of applications in fields like business, government, healthcare, and recommendation systems. A recent literature review highlighted that sentiment analysis can be conducted at different levels—document-level, sentence-level, and aspect-level—each with its own unique challenges and techniques. The review also compared various SA methods and identified state-of-the-art models, such as ChatGPT, ERNIE Bot, and PaLM. Despite significant advancements, challenges remain in handling ambiguous sentiments and improving model generalization across different domains, indicating that there is still much to explore in sentiment analysis [5].

The primary goal of this study is to evaluate the effectiveness of three distinct sentiment analysis techniques—deep learning, rule-based, and word scoring—on a large dataset of airline reviews. By assessing their performance with key metrics such as accuracy, precision, recall, and F1-score, this research provides a comparative analysis of each method's strengths and weaknesses. This work not only contributes to the growing body of sentiment analysis research but also offers practical insights for airlines in selecting the best approach for analyzing customer feedback and improving their service strategies.

The remainder of the paper is structured as follows: Section II reviews related work in sentiment analysis, particularly within the airline industry. Section III describes the methodology used in this study, including the hybrid labeling strategy and evaluation framework. Section IV provides an overview of the dataset used for experimentation, while Section V covers preprocessing and feature extraction steps. Section VI details the three sentiment analysis techniques implemented. Section VII explains the experimental setup, including train-test splits and parameter tuning. Section VIII presents the results and comparative analysis of the models. Section IX discusses the study's limitations and reflections, and Section X concludes the paper. Finally, Section XI explores future research directions, with an appendix offering supplementary information.

II. RELATED WORK

Sentiment analysis has become increasingly popular in recent years, particularly in industries like aviation, where customer satisfaction is a critical performance metric. Early studies in this field often relied on traditional machine learning models such as Support Vector Machines (SVM), Logistic Regression, and Naïve Bayes, paired with natural language processing (NLP) techniques like Bag-of-Words (BoW) and TF-IDF, to classify sentiments in large, unstructured datasets, including social media content. For example, previous research on the Twitter US Airline Sentiment dataset—known for its

size, imbalance, and multi-class nature—achieved an accuracy of 77

Building on this foundation, hybrid models have emerged to address the limitations of individual classifiers. One study combined SVM with Gradient Boosting Machines (GBM) through majority voting and achieved a 96

Traditional methods for assessing airline service quality have mostly relied on structured surveys, which often miss the nuance and immediacy of user-generated content (UGC). With the rise of social media and online reviews, researchers have turned to big data analytics—using techniques like text mining, sentiment analysis, and topic modeling—to derive insights from customer feedback. For instance, one study analyzed over 14,000 online reviews of Asian airlines, contributing to the growing body of research using unstructured data to inform customer service strategies [8].

Other research has explored sentiment analysis to evaluate service quality, starting with movie reviews and then expanding to platforms like Twitter. These studies highlight the potential of UGC as a cost-effective alternative to traditional surveys. For example, the SASQUE framework analyzed tweets to uncover strengths and weaknesses in airline services. Despite challenges in detecting sarcasm and complex phrases, such approaches offer scalable, real-time feedback mechanisms for service improvement [9].

Deep learning techniques have taken sentiment analysis even further, particularly when it comes to unstructured airline review data. Models like RNN, LSTM, GRU, CONV1D, and BERT have shown high classification accuracy, with LSTM models achieving up to 91

More recently, Rahman et al. introduced a hybrid RoBERTa-BiLSTM model, combining the contextual strengths of RoBERTa with the temporal processing capabilities of BiLSTM. Their approach achieved over 85

Another significant study focused on Twitter data related to major global airlines, using both n-gram and GloVe-based embeddings. After testing various ANN and SVM models, the ANN4 architecture emerged as the most accurate. The use of 4-gram representations notably boosted performance, and a CNN-based model further improved classification results. Additionally, association analysis helped uncover the root causes of customer sentiment, providing valuable insights for enhancing airline services [12].

Earlier research in sentiment analysis has mainly focused on leveraging machine learning techniques like Naïve Bayes and deep learning to classify customer reviews for better business decision-making. This work shows that analyzing UGC from e-commerce and social platforms can help improve product design, reduce reliance on manual feedback methods, and offer insights into consumer behavior—especially among younger, tech-savvy demographics, despite concerns about online security and system complexity [13].

BERT, a pre-trained language model from Google, has proven to be highly effective in sentiment analysis tasks—especially on airline customer reviews. It outperforms traditional machine learning models like Random Forest in

terms of accuracy, precision, recall, and F1-score. However, further improvements could come from using variants like BERT Large, ALBERT, RoBERTa, and ELECTRA, though these would require more training time and increased implementation complexity [14].

In summary, sentiment analysis has evolved from rule-based and traditional machine learning methods to advanced deep learning and hybrid approaches. The most effective models today are those that combine the contextual understanding of language models with sequential or convolutional processing. This paper builds on this progression by introducing a scalable, interpretable hybrid sentiment analysis model designed for real-world, domain-specific applications.

III. METHODOLOGY

This study employs a hybrid sentiment analysis framework, combining lexicon-based, rule-based, and deep learning-based techniques to label and evaluate customer sentiment from airline reviews. The overall pipeline consists of a structured labeling process, followed by evaluation using multiple sentiment analysis techniques.

A. Hybrid Labeling Framework

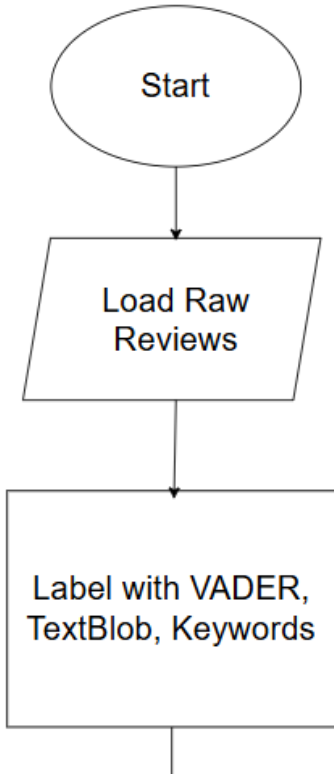


Fig. 1. Part 1: Load Raw Reviews and Apply Initial Labeling using TextBlob, VADER, and Keyword Tagging

Figure 1 this part of the labeling pipeline represents the first step, where raw airline reviews are loaded and processed through three different sentiment labeling methods: TextBlob, VADER, and a custom keyword-based approach. Each method

offers a unique perspective on sentiment—TextBlob and VADER rely on rule-based and lexicon-based evaluations, while the keyword method focuses on identifying domain-specific cues. By combining these approaches, the system lays the groundwork for identifying high-confidence labels, which are further refined through agreement checks in the next stages.

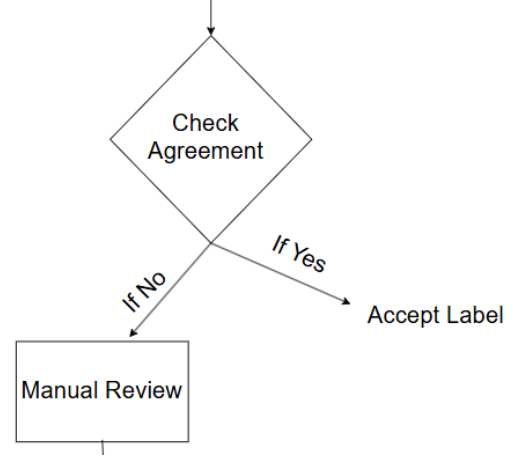


Fig. 2. Part 2: Label Agreement Check and flag uncertain reviews

Figure 2 this step represents the second phase of the hybrid labeling framework, where the outputs from the initial three sentiment labelers are compared. If all three methods agree on a sentiment label, it's accepted as a high-confidence annotation. However, if there's disagreement, the review is flagged for pseudo-labeling. This approach helps clear up ambiguities and ensures that more complex or subtle expressions of sentiment are handled accurately before moving on to the next stage of the pipeline.

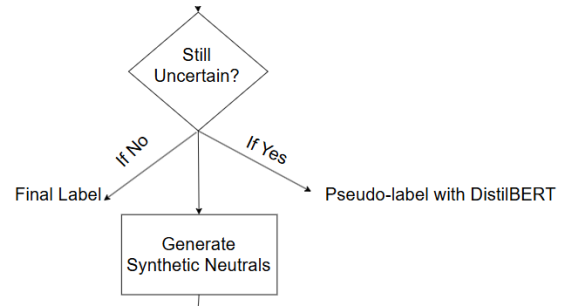


Fig. 3. Part 3: Pseudo-Labeling with distilbert-base-uncased-finetuned-sst-2-english for Ambiguous Cases and Synthetic Neutral Generation

Figure 3 this stage represents the third phase of the labeling pipeline, where reviews that remain ambiguous after pseudo-labeling are processed using a pre-trained model called distilbert-base-uncased-finetuned-sst-2-english. This model generates labels based on its understanding of the text's context. By doing so, it helps expand label coverage without depending solely on manual annotation. Additionally, to balance the dataset and reduce sentiment bias, synthetic neutral reviews are created and added to the final labeled data.

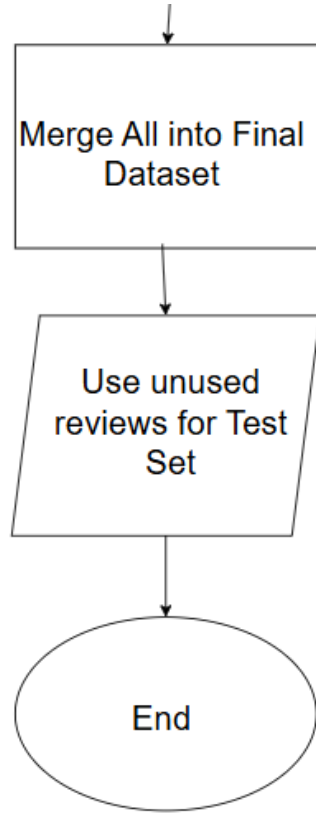


Fig. 4. Part 4: Merge All Labels into Final Training Set and Reserve Remaining Reviews for Test Set

Figure 4 this describes the final phase of the hybrid labeling process. At this point, all confidently labeled reviews—whether they were fully agreed upon or flagged for uncertainty and later pseudo-labeled—are gathered to create the validation dataset. Any remaining unused reviews are set aside as an independent test set. Reviews with unanimous (3/3) sentiment agreement are given priority, and when that’s not possible, pseudo-labeled reviews are used instead. This data split ensures that model evaluation is based on samples not seen during training, providing a trustworthy assessment of how well the model generalizes and performs overall.

We started with 22,098 reviews scraped from airlinequality.com. These reviews were then processed using three initial labeling techniques: TextBlob, VADER, and keyword/topic-based sentiment tagging. If all three methods agreed on the sentiment label, it was accepted right away. When there was disagreement, the review was flagged for pseudo-labeling. For reviews that remained unclear, we used pseudo-labeling with the distilbert-base-uncased-finetuned-sst-2-english model to finalize the labels for those uncertain reviews.

B. Evaluation Methodology

Three sentiment classification methods were evaluated:

- **Lexicon-Based:** AFINN-based word scoring.
- **Rule-Based:** SentiWordNet with polarity aggregation.
- **Deep Learning-Based:** cardiffnlp/twitter-xlm-roberta-base-sentiment.

Each model was applied to the test set, and performance was measured against the validation set using standard metrics: Accuracy, Precision, Recall, and F1-score.

C. Validation Set

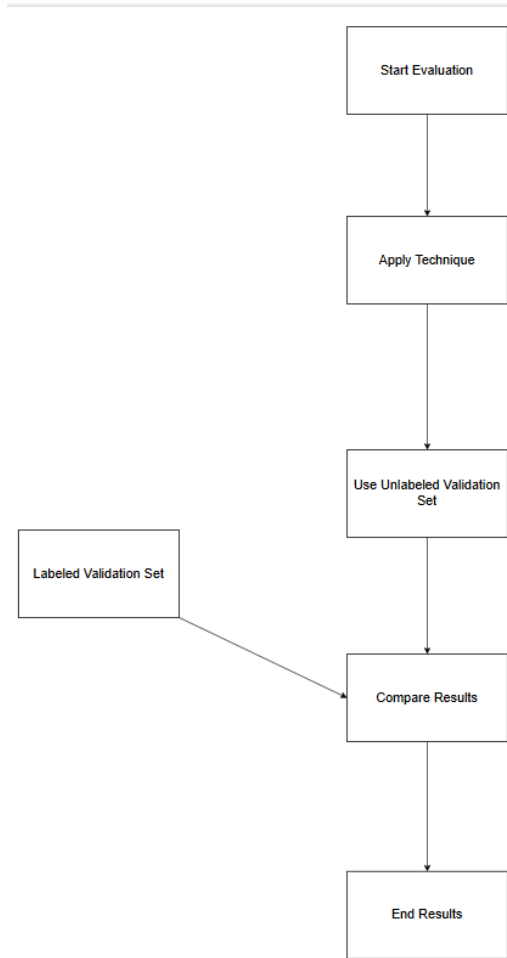


Fig. 5. Validation Set Construction Process

The final labeled validation set is composed of four sources:

- Automatically labeled samples from unanimous agreement.
- Flag uncertain reviews.
- Pseudo-labeled flagged reviews using distilbert-base-uncased-finetuned-sst-2-english.
- Synthetic neutral reviews generated using GPT-4 to balance the dataset.

This validation set serves as the ground truth for evaluating sentiment analysis techniques.

D. Test Set

The remaining labeled reviews are set aside as the test set. These reviews are excluded from all labeling and model training steps to ensure an unbiased evaluation of how well the sentiment analysis techniques generalize. Reviews with full (3/3) sentiment agreement are prioritized; if none are available,

those with partial (2/3) agreement are used, followed by pseudo-labeled samples if necessary. This approach ensures a balanced and representative distribution between the validation and test sets.

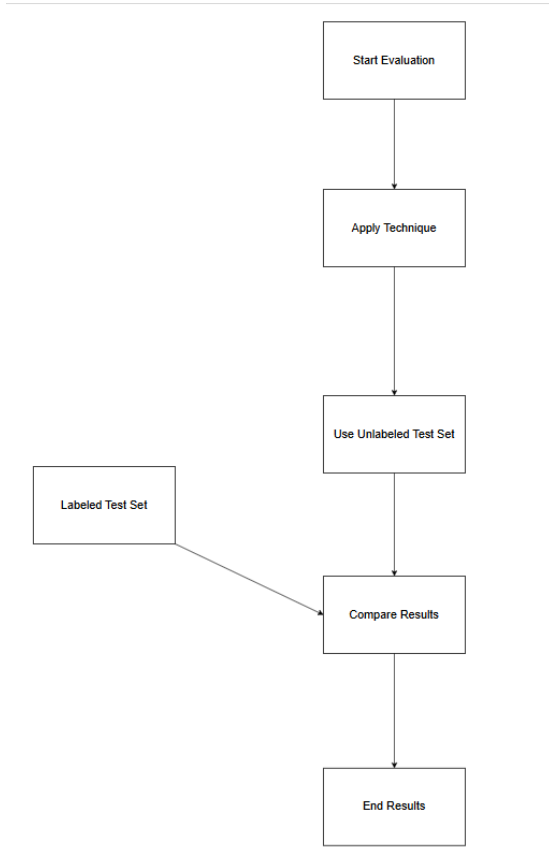


Fig. 6. Test Set Allocation Process

IV. DATA

The dataset for this project was created by web scraping user reviews from the Skytrax website¹, a well-known platform for airline and airport feedback. In total, 22,098 reviews were collected, spanning various airlines, airports, and routes. These reviews include both qualitative and quantitative feedback on customer experiences.

Each review contains the following fields:

- **Reviewer Name:** The name of the reviewer.
- **Date of Submission:** The date when the review was submitted.
- **Star Rating:** A numeric rating from 1 to 10 provided by the reviewer to express overall satisfaction.
- **Review Text:** A textual description of the reviewer's experience.
- **Sentiment Labels:** These are automatically assigned based on the sentiment of the review (Positive, Neutral, Negative).

¹<https://www.airlinequality.com>

SAMPLE REVIEWS

Here are some samples from the dataset, representing different sentiments and types of review content. Each example highlights a distinct sentiment classification (Negative, Neutral, and Positive), illustrating the variety of expressions and contexts that the models need to handle.

- **Airline:** Petroleum Air Services
Reviewer: Diogo Freitas
Date: 2018-10-13
Rating: 10.0
Review Text: "We flew to Cairo from our Sharm el-Sheikh resort as many others did in order to tour the capital city of Egypt. The aircraft was very comfortable and seemed quite new. They offer great catering for the one hour flight. Couldn't fault it."
Sentiment: Positive
- **Airport:** Abu Dhabi
Reviewer: Gary Colburn
Date: 2024-10-18
Rating: 8.0
Review Text: "Super modern airport, bright, shiny and very clean. Security control was reasonably quick on departure and arrival and I found the staff professional (although they don't smile a lot). Bonus points for having showers after security—I took one and it was marvellous. I wasn't sold on Abu Dhabi itself, but the airport is certainly one of the best I've experienced in recent times."
Sentiment: Positive
- **Airport:** Naples Municipal
Reviewer: Johannes Schunter
Date: 2012-10-09
Rating: 10.0
Review Text: "The airport has a quick check-in counter right outside where you arrive with the taxi. This led to a check-in time for our Delta flight (including one checked-in luggage) of just minutes after I got out of the taxi. Security was fast too—never experienced such a comfortable on-boarding on any flight."
Sentiment: Positive
- **Airline:** Virgin Atlantic
Reviewer: Anonymous
Date: Unknown
Rating: N/A
Review Text: "Virgin Atlantic's flight was punctual, taking off and landing on the scheduled times."
Sentiment: Neutral
- **Airline:** Air France
Reviewer: Anonymous
Date: Unknown
Rating: N/A
Review Text: "Air France's flight was punctual and the check-in process was efficient. The legroom was a bit limited."
Sentiment: Neutral
- **Airline:** Turkish Airlines

Reviewer: Anonymous

Date: Unknown

Rating: N/A

Review Text: "The Turkish Airlines flight was typical. The staff was professional, the food was decent, and the flight was on schedule."

Sentiment: Neutral

- **Airline:** Blu-express

Reviewer: G Toppazze

Date: 2016-09-01

Rating: 2.0

Review Text: "Heraklion to Milan Bergamo. Flight almost 4 hrs late, no assistance, no information. Horrendous experience and no one from Blu-express is showing up. Stuck in the airport and no clue what we'll be doing when arrived at destination since all transports will be over by that time."

Sentiment: Negative

- **Airline:** Brussels Airlines

Reviewer: E Darnezzì

Date: 2025-01-31

Rating: 1.0

Review Text: "Unfortunately my experience with Brussels Airlines has been horrible. The rudeness of the ground staff combined with the delayed flights made my experience with this airline truly tiring. I tried to complain multiple times but their customer service was simply incompetent."

Sentiment: Negative

- **Airline:** Nouvelair

Reviewer: Hamdi Chahdi

Date: 2023-05-01

Rating: 1.0

Review Text: "I had a terrible experience with Nouvelair on the flight BJ. I booked a flight for 5AM on the 1st. I received an email—they rescheduled the flight to be 5PM without giving any options. After the flight, a 4 hours 30 minutes delay—so instead of arriving at 7AM, I arrived at 5PM. Better to avoid this company whatever the cost."

Sentiment: Negative

A. Dataset Splits and Labeling

From the scraped dataset, 12,000 reviews were selected for labeling, while the remaining 3,000 reviews were set aside as an unlabeled test set. The labeled reviews were processed using the hybrid labeling strategy described in the Methodology section, resulting in a balanced dataset with 4,000 positive, 4,000 neutral, and 4,000 negative reviews.

Final dataset structure:

- **Labeled Validation Set:** 12,000 reviews (4,000 per sentiment class)
- **Unlabeled Test Set:** 3,000 reviews reserved for evaluating model generalization

V. PREPROCESSING AND FEATURE EXTRACTION

Before applying sentiment analysis techniques, the raw airline review data needed thorough pre-processing to eliminate noise and ensure consistency. This phase involved cleaning the text and preparing it for feature extraction and model input.

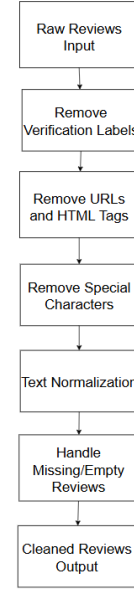


Fig. 7. Preprocessing Workflow for Airline Review Text

Figure 7 outlines the step-by-step preprocessing pipeline used in the study. The process starts with raw text ingestion and moves through several cleaning stages, including label stripping, special character removal, normalization, and text validation. This structured approach ensures that only relevant, high-quality review content is kept for the next steps of sentiment analysis and feature extraction.

A. Preprocessing Steps

The following operations were applied to each review:

- **Verification Label Removal:** Introductory phrases such as "Verified Review" or "Unverified" were removed using regular expressions, as they carried no sentiment value.
- **URL and HTML Tag Removal:** Embedded hyperlinks and HTML artifacts were stripped out to ensure only the review text remained.
- **Special Character Cleaning:** Non-alphanumeric symbols (e.g., #,) were removed unless they were part of valid punctuation or relevant to sentiment expression.
- **Text Normalization:** All text was converted to lowercase, and extra whitespace was trimmed to standardize the formatting, which is crucial for consistent tokenization.
- **Missing Text Handling:** Empty or invalid reviews were excluded from the dataset to preserve its quality and maintain a focus on genuine feedback.

After preprocessing, the cleaned dataset was stored as `cleaned_skytrax_reviews.csv`.

B. Feature Extraction

The cleaned text data was used to extract features in different ways depending on the sentiment analysis technique:

- **Rule-Based Method:** This approach uses predefined linguistic rules and sentiment lexicons to classify text. For example, TextBlob determines sentiment polarity based on established sentiment scores in its lexicon. Additionally, a custom keyword-based sentiment dictionary was created, which assigns sentiment values based on the presence and frequency of sentiment-rich terms. The rule-based model doesn't require training data and is easy to interpret, but its effectiveness largely depends on the coverage and quality of the lexicon, as well as how well the rules capture the nuances of the text.
- **Word Score-Based Method:** AFINN sentiment scores were calculated by mapping each token in the review to its corresponding sentiment value and then aggregating the results. This method is especially effective at capturing the sentiment of individual words and phrases within the reviews.
- **Deep Learning Method:** For the cardiffnlp/twitter-xlm-roberta-base-sentiment model, each review was broken down into subword units and input into a pre-trained RoBERTa transformer, which generates contextual embeddings. These embeddings are then used to predict the sentiment class probabilities (positive, neutral, or negative), considering the surrounding context and nuances within the review.

These different feature extraction approaches enabled us to compare how simpler, rule-based methods like TextBlob and AFINN perform versus more advanced transformer-based models like XLM-RoBERTa. This comparison is crucial for understanding the strengths and weaknesses of various sentiment analysis techniques, especially in the context of airline reviews, where sentiment can be nuanced, diverse, and complex.

VI. METHODS

This study uses three different sentiment analysis techniques to evaluate airline reviews: Word-Score Sentiment Scoring with AFINN, Rule-Based Sentiment Analysis with SentiWordNet, and Deep Learning-Based Sentiment Analysis using cardiffnlp/twitter-xlm-roberta-base-sentiment. These methods were selected to offer a well-rounded analysis, combining traditional lexicon-based approaches with modern machine learning techniques. Each method has its own strengths and limitations, which are outlined below.

A. Overview of Sentiment Analysis Methods: Strengths and Weaknesses

This study evaluates three distinct sentiment analysis techniques: lexicon-based (AFINN), rule-based (SentiWordNet), and deep learning-based (XLM-RoBERTa). Each approach has its own strengths and limitations, depending on the nature of the text and the computational resources at hand. The

following sections provide a summary of the advantages and drawbacks of each method.

a) Lexicon-Based Method (AFINN)

AFINN is a straightforward yet effective technique that assigns sentiment scores to individual words based on a predefined lexicon. Its main advantage is its computational efficiency and ease of implementation, making it ideal for large-scale, real-time applications. However, its simplicity also limits its effectiveness—it doesn't account for the context in which words are used, and struggles with complex sentence structures, sarcasm, or words with multiple meanings.

b) Rule-Based Method (SentiWordNet)

SentiWordNet builds on the lexicon-based approach by incorporating rule-based logic through WordNet synsets, which helps better capture context and semantic relationships between words. This enhancement allows for more accurate sentiment scoring by analyzing not only individual words but also their meanings in different contexts. However, rule-based methods still face challenges with highly implicit sentiment, idiomatic expressions, or domain-specific language that falls outside the predefined rules and lexicons.

c) Deep Learning-Based Method (XLM-RoBERTa)

The deep learning approach uses the XLM-RoBERTa model, a powerful transformer-based architecture pre-trained on a large multilingual corpus. This method excels at capturing nuanced sentiment and understanding the broader context of reviews, particularly in longer or more complex texts. It is especially effective at detecting subtle emotional tones, irony, or mixed sentiments. However, these advantages come with the trade-off of higher computational requirements, longer processing times, and reduced interpretability compared to lexicon or rule-based approaches.

Overall, these three methods strike a balance between interpretability, computational cost, and sentiment detection capability. This paper compares their performance using four key evaluation metrics—accuracy, precision, recall, and F1-score—along with a confusion matrix to assess how effectively each method classifies sentiment across different categories.

VII. EXPERIMENTS

A. Computer Specifications

The sentiment analysis experiments were conducted on the following system:

- **Processor (CPU):** AMD Ryzen 7 3700X (8 cores, 16 threads)
- **Graphics Card (GPU):** NVIDIA GeForce RTX 2070 Super Titan
- **RAM:** 32GB DDR4 (G.Skill Ripjaws V Series)
- **Storage:** 500GB WD_BLACK NVMe SSD, 2TB Seagate Barracuda HDD
- **Motherboard:** ASUS AM4 TUF Gaming X570-Plus (WiFi)
- **CPU Cooler:** Corsair H100i RGB Platinum
- **Power Supply:** Corsair RM850x
- **Case:** CORSAIR 275R (White)
- **Operating System:** Windows 10, 64-bit

- **Python Version:** Python 3.11.9

These system specifications provided the necessary computational resources to efficiently run sentiment analysis models, especially the deep learning approach, which required significant GPU acceleration. The AMD Ryzen 7 3700X (8 cores, 16 threads) facilitated smooth multi-threaded processing, while the NVIDIA GeForce RTX 2070 Super Titan (8GB VRAM) enabled fast model training and inference for transformer-based models.

With 32GB of DDR4 RAM, the system was capable of handling large datasets and multiple concurrent processes without performance issues. The 500GB NVMe SSD ensured fast data access and reduced I/O bottlenecks, while the 2TB HDD offered ample storage space for datasets, model outputs, and checkpoints.

A Corsair RM850x PSU provided stable power delivery during extended training sessions, and the H100i RGB Platinum cooler kept the system's temperature in check, preventing thermal throttling. The system ran on Windows 10 (64-bit) with Python 3.11.9, ensuring compatibility with essential libraries like TensorFlow, PyTorch, and Scikit-learn, which were crucial for implementing and evaluating the sentiment analysis models.

B. Train-Test Division

The dataset was divided into training and test sets using an 80-20 split, with 80% of the data used for training and 20% reserved for testing. This approach ensured that the models were trained on a representative subset of the reviews, while their performance was evaluated on a separate set of unseen data. This partitioning strategy is essential for assessing the models' ability to generalize and their robustness when applied to new, previously unseen reviews.

C. Data Preprocessing

Prior to the application of sentiment analysis methods, several preprocessing steps were carried out to clean and standardize the input data. These steps included:

- **Text Cleaning:** URLs, HTML tags, non-alphanumeric characters, and extra spaces were removed from the reviews. Additionally, all text was converted to lowercase to maintain consistency and prevent case sensitivity from affecting sentiment analysis.
- **Tokenization and POS Tagging:** Each review was tokenized, and part-of-speech (POS) tagging was performed. This was especially important for methods like Rule-Based Sentiment Analysis using SentiWordNet, where part-of-speech tagging helps in determining the sentiment of a word based on its context in the sentence.
- **Stopword Removal:** Common stopwords were removed, except for negations (e.g., "not", "never"), as they play an important role in sentiment analysis by altering the meaning of other words.

D. Model Implementation and Hyperparameters

In this study, three distinct sentiment analysis methods were implemented: Word-Score Sentiment Scoring using AFINN, Rule-Based Sentiment Analysis with SentiWordNet, and Deep Learning-Based Sentiment Analysis using the pre-trained cardiffnlp/twitter-xlm-roberta-base-sentiment model. Each method offers a unique approach to sentiment classification, and the following sections provide detailed explanations of how each method works.

1) Word-Score Sentiment Scoring w/ AFINN

The AFINN model is a word-level sentiment analyzer that assigns each word a sentiment score. For this method:

- **Model:** The AFINN model uses a lexicon where each word is associated with a sentiment score. The overall sentiment score for a review is obtained by summing the individual word scores.
- **Hyperparameters:**
 - Positive and negative keywords (e.g., "amazing", "horrible") were manually defined and adjusted to boost or penalize the sentiment score accordingly.
 - Neutral words or phrases (e.g., "okay", "average") were included with small penalties or boosts based on their influence on the sentiment score.
- **Keyword Adjustments:**
 - Strong positive and negative words were given higher boosts (e.g., "amazing" adds +1.1).
 - Soft neutral words or phrases were adjusted with a smaller penalty or boost based on their influence.
- **Sentiment Thresholds:** The final sentiment classification is determined based on the score thresholds:
 - Positive sentiment: score > 2.0
 - Negative sentiment: score < -0.9
 - Neutral sentiment: scores in between.

2) Rule-Based Sentiment Analysis w/ SentiWordNet

The Rule-Based method leverages the SentiWordNet lexicon to calculate sentiment scores based on the part-of-speech tagging of words in the review. The detailed adjustments are as follows:

- **Model:** The SentiWordNet lexicon assigns sentiment scores to words based on their part-of-speech tags. Each word's sentiment score is calculated by referencing the appropriate word sense from the lexicon.
- **Hyperparameters:**
 - Part-of-speech tagging was used to determine how words contribute to the overall sentiment based on their context (e.g., noun, verb, etc.).
 - A rule-based system was applied to resolve conflicting sentiments in a review, where a weighted average score or majority rule is used.
- **Negation Handling:** Negations (e.g., "not", "never") were handled by flipping the sentiment score of words they negated, ensuring that sentiment inversion was accurately reflected in the analysis.

- **Intensifier Handling:** The presence of intensifiers (e.g., "very", "extremely") boosted the sentiment score of the words they modified. A list of common intensifiers was used to scale sentiment scores.
- **Keyword Boosting:** Strong sentiment words (e.g., "amazing", "horrible") were used to adjust the sentiment score, boosting or penalizing the overall sentiment when detected in a review.

3) Deep Learning-Based Sentiment Analysis w/ *cardiffnlp/twitter-xlm-roberta-base-sentiment*

For deep learning-based sentiment analysis, we used the pre-trained *cardiffnlp/twitter-xlm-roberta-base-sentiment* model. This transformer-based model was fine-tuned for sentiment classification on Twitter data, making it effective for general sentiment tasks. The adjustments were as follows:

- **Model:** The *cardiffnlp/twitter-xlm-roberta-base-sentiment* pre-trained model was used from Hugging Face's Transformers library.
- **Hyperparameters:**
 - The model was run with a batch size of 32, utilizing GPU acceleration for faster processing.
 - The default pre-trained model parameters were used to ensure reliable performance.
- **Thresholding:** To ensure reliable predictions, sentiment labels with low confidence (i.e., confidence below 0.7) were adjusted to neutral. This was done to prevent misclassification due to low model confidence.
- **Keyword Override and Boosting:** Strong sentiment keywords (e.g., "excellent", "horrible") were used to override the model's prediction if detected in the review text. Additionally, positive sentiment keywords (e.g., "good", "pleasant") were used to boost or adjust the final sentiment classification.

E. Evaluation Setup

To evaluate the performance of each model, we used several key metrics:

- **Accuracy:** The percentage of correct sentiment predictions (positive, negative, or neutral) compared to the ground truth.
- **Precision, Recall, F1-Score:** These metrics were computed for each sentiment class (positive, negative, and neutral) to assess the models' effectiveness at correctly identifying each class.

Each model was evaluated using distinct validation and test sets to ensure robust performance and prevent overfitting. This approach allows for tuning the models on the validation set while then assessing their final performance on an unseen test set, helping to evaluate the models' ability to generalize to new, unseen data.

F. Runtime Analysis

The time required to run sentiment analysis on both the validation and test sets varied significantly between the models, reflecting the differences in their computational complexity.

Below is a breakdown of the runtime for each model on both datasets:

- **Word-Score Sentiment Scoring (AFINN) and Rule-Based Sentiment Analysis (SentiWordNet):** Both the Word-Score Sentiment Scoring method (AFINN) and the Rule-Based Sentiment Analysis method (SentiWordNet) were highly efficient in terms of processing time. These methods are based on lexicons and do not require training or deep learning techniques, resulting in very fast execution. The time taken to analyze both the validation and test sets was under 30 seconds for each model. This efficiency makes lexicon-based approaches suitable for large datasets where speed is a crucial factor.

- **Deep Learning-Based Sentiment Analysis (*cardiffnlp/twitter-xlm-roberta-base-sentiment*):**

In contrast, the deep learning-based sentiment analysis using the pre-trained *cardiffnlp/twitter-xlm-roberta-base-sentiment* model required significantly more time due to its complexity. For the validation set, the model took approximately 13 minutes to complete the analysis. The longer runtime is attributed to the model's size and the need for processing each review through a complex neural network, which involves multiple layers of computation. On the test set, the analysis took about 6 minutes, reflecting the model's relatively higher efficiency after processing the validation set. Despite the longer processing time, deep learning methods offer superior accuracy and can capture more nuanced sentiment, which makes the trade-off in runtime worthwhile in contexts where precision is more important than speed.

These runtime results highlight the trade-off between the simplicity and speed of lexicon-based methods and the accuracy and computational cost of deep learning models. While lexicon-based methods deliver near-instantaneous results, deep learning models—though slower—are capable of capturing more complex sentiment. They offer better performance in terms of overall accuracy, particularly when dealing with large and diverse datasets, such as airline reviews, where the sentiment can be nuanced and varied.

G. Summary of Experiment Setup

- **Data Split:** 80% training, 20% testing.
- **Preprocessing:** Text cleaning, tokenization, stopword removal, and POS tagging.
- **Hyperparameters:** Default settings for AFINN, SentiWordNet, and deep learning models, with keyword boosting and thresholding applied where applicable.
- **Cross-Validation:** 5-fold cross-validation for all models.
- **Evaluation Metrics:** Accuracy, precision, recall, and F1-score for sentiment classification.

This experimental setup provided a consistent and objective framework for evaluating each sentiment analysis method, ensuring a thorough and fair comparison of their effectiveness across a diverse and representative set of airline reviews.

VIII. RESULTS

In this section, we will evaluate the performance of the deep learning sentiment analysis model by looking at several key metrics: accuracy, precision, recall, and F1-score. These metrics provide a comprehensive assessment of the model's performance. Accuracy measures overall correctness, while precision focuses on the model's ability to correctly identify each sentiment class. Recall, on the other hand, evaluates how sensitive the model is to detecting all relevant instances of each class. The F1-score balances both precision and recall, giving us a clearer picture of the model's performance, especially in cases where the data may be imbalanced.

A. Deep Learning Model Results

The deep learning model (cardiffnlp/twitter-xlm-roberta-base-sentiment) was evaluated on the test set. Below are the results, including the classification report, confusion matrix, and evaluation metrics.

1) Confusion Matrix (Test Set)

The confusion matrix for the deep learning model offers a detailed breakdown of how the true sentiment classes compare with the predicted classes. By showing the number of true positives, false positives, true negatives, and false negatives for each sentiment category (Negative, Neutral, and Positive), it allows us to identify where the model is making misclassifications. This insight helps in understanding which sentiment classes the model has difficulty distinguishing, guiding us toward potential improvements in model training, feature selection, or post-processing techniques to enhance classification accuracy.

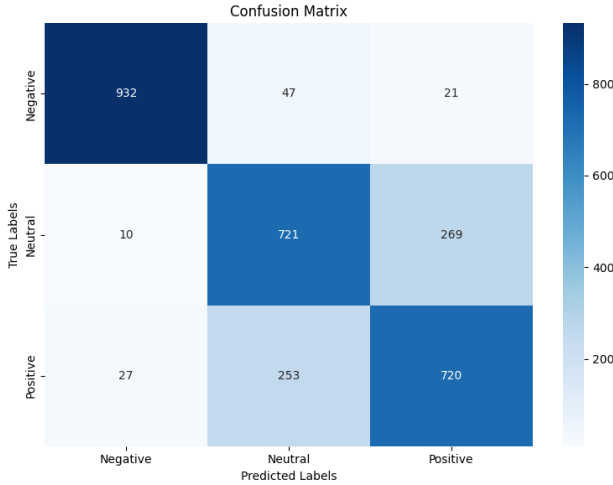


Fig. 8. Confusion Matrix for Deep Learning Model (Test Set)

For the Negative class, most of the 1000 negative reviews were correctly classified as negative (932), with a small portion misclassified as neutral (47) or positive (21). In terms of Neutral sentiment, 721 of the 1000 neutral reviews were correctly classified, while the remaining reviews were misclassified as positive (269) and negative (10). For the Positive class,

720 positive reviews were correctly identified, but some were misclassified as neutral (253) and negative (27). Overall, while the model performs well in identifying negative sentiment, there are some issues with misclassifications between the neutral and positive categories, showing room for improvement in handling these sentiments.

2) Precision, Recall, and F1-Score per Class

The following graphs show the precision, recall, and F1-score for each sentiment class (Negative, Neutral, Positive) from the deep learning model.

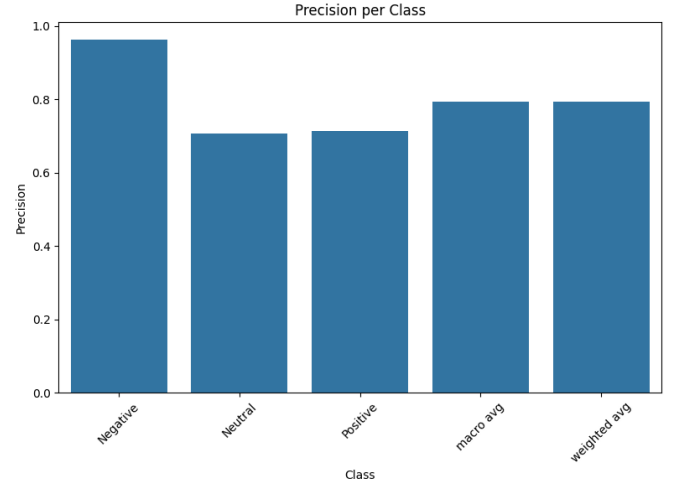


Fig. 9. Precision per Class for Deep Learning Model

In Figure 9, the model demonstrates excellent precision for Negative sentiment, with scores near 1.0, indicating minimal misclassification. For Neutral and Positive sentiments, precision ranges from 0.6 to 0.8, showing more frequent false positives. Both macro and weighted average precision scores are around 0.8, reflecting balanced overall performance. While the model is reliable in detecting Negative sentiment, there's room for improvement in Neutral and Positive precision.

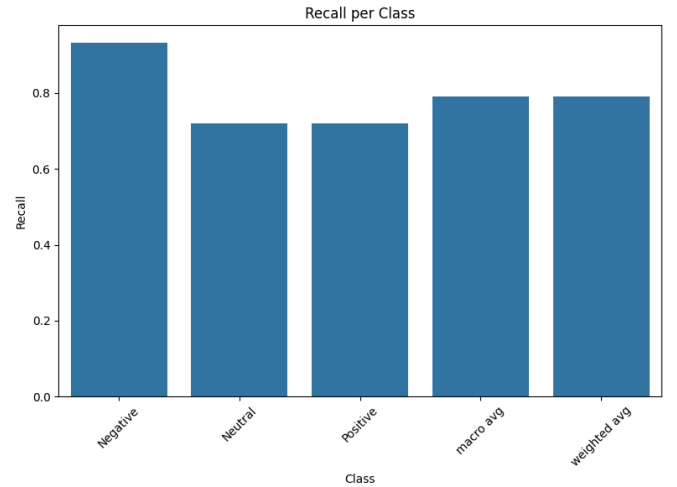


Fig. 10. Recall per Class for Deep Learning Model

In Figure 10, the model shows strong recall, excelling in detecting Negative sentiment with a score close to 1.0. For Neutral and Positive sentiments, recall ranges between 0.6 and 0.8, indicating some missed instances. Both macro and weighted average recall scores are around 0.8, reflecting consistent performance. While the model excels at detecting Negative sentiment, there's room for improvement in capturing Neutral and Positive sentiments more comprehensively.

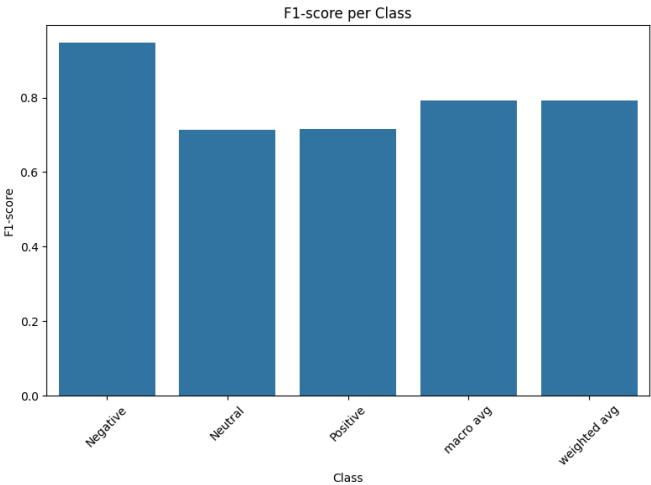


Fig. 11. F1-Score per Class for Deep Learning Model

In Figure 11, the model achieves exceptional recall for Negative sentiment, near 1.0, effectively identifying most true Negative instances. Recall for Neutral and Positive classes is moderate, between 0.6 and 0.8, indicating some missed true examples. Overall, the macro and weighted average recall scores around 0.8 reflect consistent performance, with room for improvement in reliably identifying Neutral and Positive sentiments.

3) *Precision, Recall, and F1-Score Comparison*

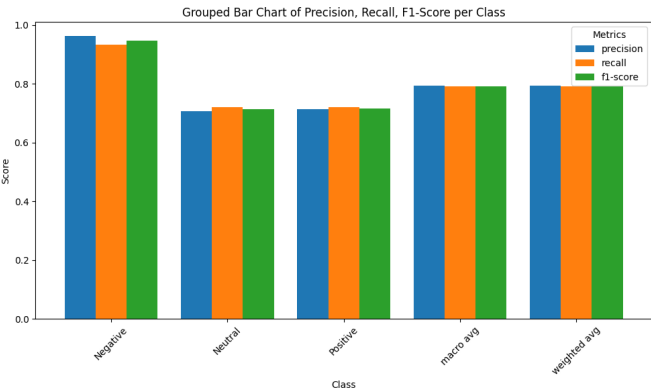


Fig. 12. Precision, Recall, and F1-Score Comparison for Deep Learning Model

In Figure 12, a grouped bar chart compares the precision, recall, and F1-scores for each sentiment class—Negative,

Neutral, and Positive. This visual comparison provides an overview of the model’s performance across these key metrics, allowing us to see where the model excels (e.g., Negative sentiment) and where there may be room for improvement (e.g., Neutral and Positive classes).

4) *Accuracy*

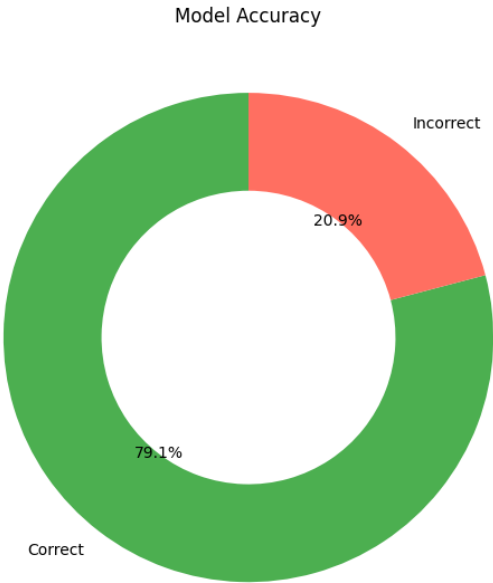


Fig. 13. Overall Accuracy of Deep Learning Sentiment Model

In Figure 13, the deep learning model (cardiffnlp/twitter-xlm-roberta-base-sentiment) achieved an accuracy of 79.1% on the test set, demonstrating strong performance in classifying sentiment across a range of airline reviews. This high accuracy highlights the model’s effectiveness in capturing nuanced sentiment and handling complex tasks in unstructured customer feedback data.

5) *Comparative Analysis*

The model excels at identifying Negative sentiment, showing high precision and recall. However, it struggles with Neutral sentiment, with lower precision and recall, often misclassifying neutral reviews as positive. For Positive sentiment, the performance is relatively balanced, though there is still room for improvement in distinguishing it from neutral sentiment.

In summary, the deep learning model performs well in classifying negative sentiment, but its ability to identify neutral and positive sentiments is more moderate. The confusion matrix shows that neutral sentiments are frequently misclassified, indicating that further fine-tuning or the inclusion of additional training data could help enhance the model’s accuracy in detecting neutral reviews.

B. *Rule-based Method Results*

The rule-based sentiment analysis model was evaluated on the test set. Below are the results, including the classification report, confusion matrix, and evaluation metrics.

1) Confusion Matrix (Test Set)

The confusion matrix for the rule-based model provides a detailed breakdown of correct and incorrect classifications for each sentiment class.

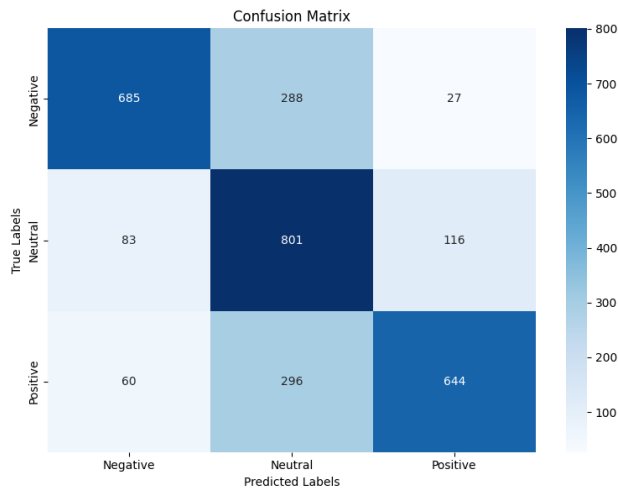


Fig. 14. Confusion Matrix for Rule-based Method (Test Set)

For the Negative class, out of the 1000 negative reviews, 685 were correctly classified as negative, while 288 were misclassified as neutral and 27 as positive. In the case of the Neutral class, 801 of the 1000 neutral reviews were correctly identified, with 83 misclassified as negative and 116 as positive. Regarding the Positive class, 644 positive reviews were correctly classified, while 60 were misclassified as negative and 296 as neutral. This shows that while the model performs well in identifying neutral reviews, there are notable misclassifications, especially with positive and neutral sentiments, indicating areas for improvement in classification accuracy.

2) Precision, Recall, and F1-Score per Class

The following graphs show the precision, recall, and F1-score for each sentiment class (Negative, Neutral, Positive) from the rule-based model.

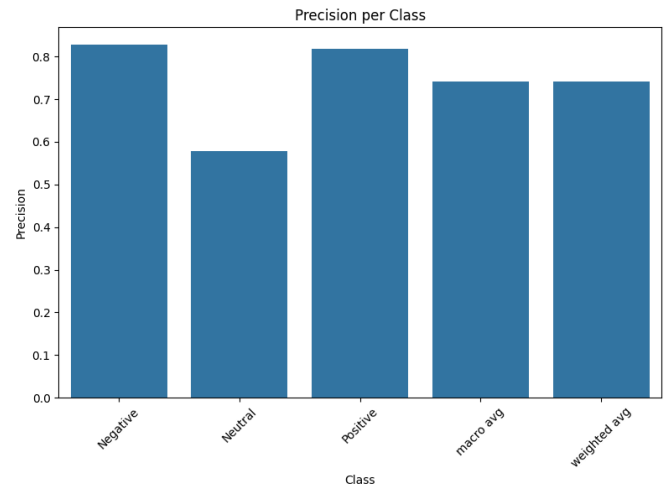


Fig. 15. Precision per Class for Rule-based Method

In Figure 15, the rule-based model shows strong precision for Negative and Positive sentiments, with scores above 0.9, indicating high accuracy and few false positives. However, precision drops below 0.6 for Neutral sentiment, suggesting frequent misclassification. Despite this, the overall performance is solid, with both macro and weighted average precision scores above 0.7, demonstrating general reliability.

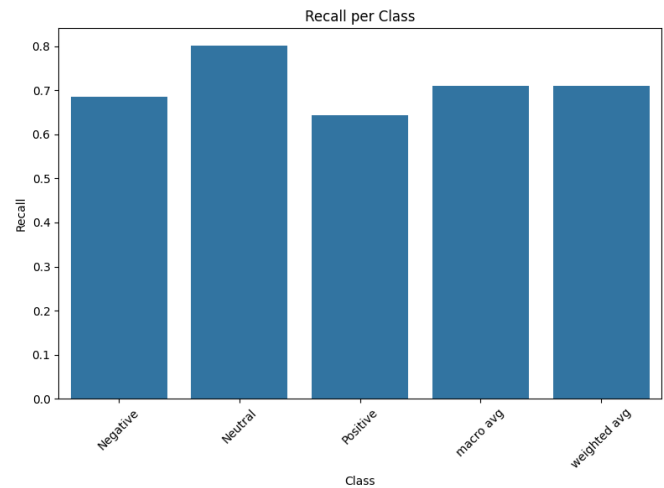


Fig. 16. Recall per Class for Rule-based Method

In Figure 16, the rule-based model shows varied recall performance. It performs well with Neutral sentiment, achieving a recall of around 0.8, but is less effective with Negative and Positive sentiments, with recall just below 0.7 for both. The macro and weighted average recall scores are just above 0.7, indicating balanced performance with room for improvement.

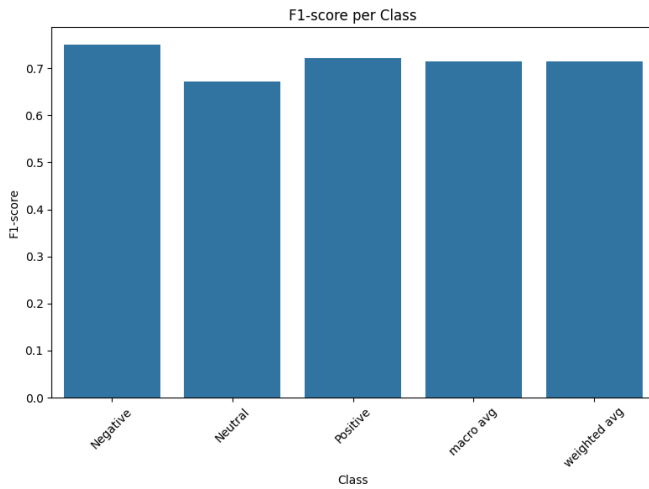


Fig. 17. F1-Score per Class for Rule-based Method

In Figure 17, the F1-scores for the rule-based model show solid performance for Negative and Positive sentiments, with scores above 0.7, indicating a good balance between precision and recall. However, Neutral sentiment lags with an F1-score below 0.7, indicating difficulties in detection and classification. Overall, the macro and weighted average F1-scores are both above 0.7, reflecting strong performance.

3) Precision, Recall, and F1-Score Comparison

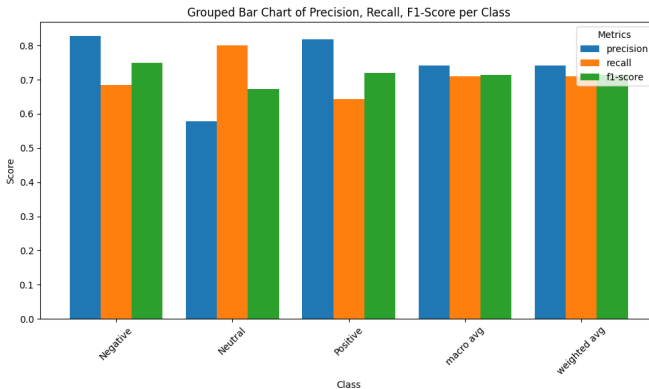


Fig. 18. Precision, Recall, and F1-Score Comparison for Rule-based Method

In Figure 18, the grouped bar chart compares precision, recall, and F1-scores for the rule-based method. The Negative and Positive classes show high precision and F1-scores (above 0.7), but lower recall, indicating some true instances are missed. The Neutral class performs weaker, with precision under 0.6 and F1-score below 0.7, despite good recall around 0.8, suggesting it struggles with precision. Overall, the macro and weighted average scores are above 0.7, indicating strong performance with room for improvement in Neutral sentiment classification.

4) Accuracy

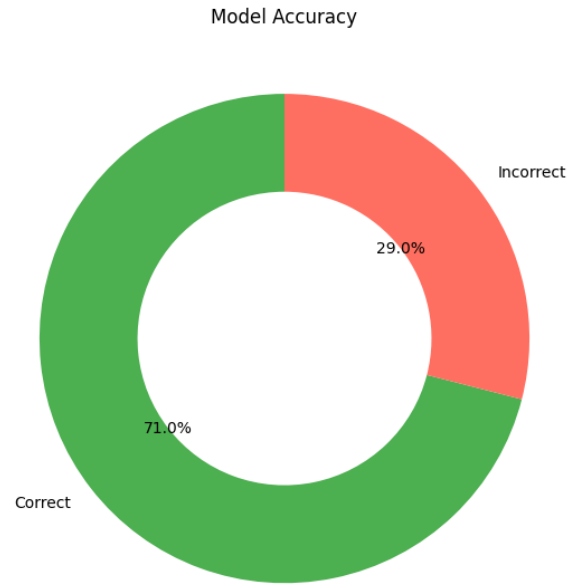


Fig. 19. Overall Accuracy of Rule-Based Sentiment Method

In Figure 19, the rule-based sentiment classification approach achieves an accuracy of 71.0%. While less context-sensitive than deep learning models, it offers a solid baseline, highlighting the strengths and limitations of traditional rule-based methods, especially for nuanced data like airline reviews.

5) Comparative Analysis

The rule-based model performs reasonably well in identifying Negative sentiment, though there are some misclassifications into the Neutral category. It has significant difficulties with Neutral sentiment, as evidenced by frequent misclassifications. For Positive sentiment, the model shows fairly good precision, but recall could still be improved, as many positive reviews are misclassified as neutral.

In summary, the rule-based model performs decently overall, but it struggles with neutral sentiment classification. Compared to the deep learning model, which achieved an accuracy of 79.1%, the rule-based model's accuracy of 71% is lower. The rule-based method would benefit from improvements in distinguishing neutral sentiment, as it tends to misclassify these reviews as either positive or negative. Despite its limitations, it still provides a useful baseline for sentiment analysis, especially in scenarios where computational efficiency is crucial.

C. Word Scoring Model Results

The word scoring model was evaluated on the test set. The results below include graphical representations of the classification report, confusion matrix, and evaluation metrics.

1) Confusion Matrix (Test Set)

The confusion matrix visually breaks down the model's predictions against the true labels.

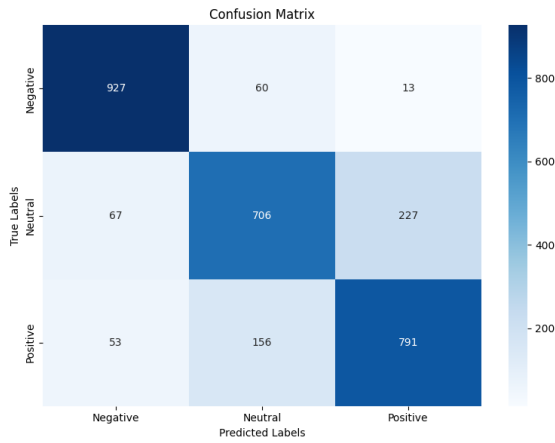


Fig. 20. Confusion Matrix for Word Scoring Model (Test Set)

For the Negative class, 653 reviews were correctly classified, with 278 misclassified as neutral and 69 as positive. In the Neutral class, 768 reviews were correctly identified, while 45 were misclassified as negative and 187 as positive. For the Positive class, 688 reviews were correctly classified, with 61 misclassified as negative and 251 as neutral. These results highlight areas where misclassifications occur, particularly with neutral and positive sentiments, suggesting potential improvements for the model in distinguishing between these classes.

2) Precision, Recall, and F1-Score per Class

Bar plots below highlight performance per sentiment category.

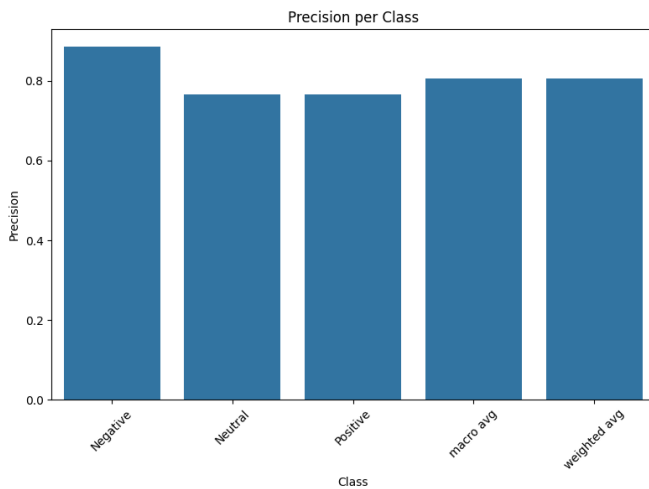


Fig. 21. Precision per Class for Word Scoring Model

In Figure 21, the word scoring model (AFINN) shows strong precision for Negative sentiment (above 0.8), with

minimal false positives. Precision for Neutral and Positive sentiments is just under 0.8, indicating some misclassification of these sentiments as negative. Overall, the model performs consistently across all classes, with solid precision around 0.8.

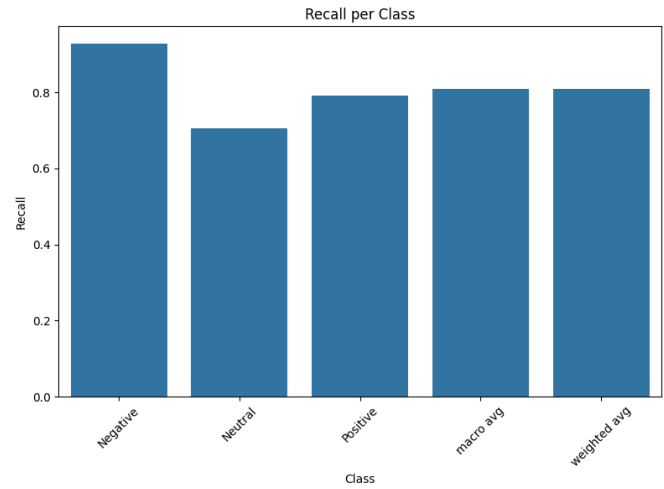


Fig. 22. Recall per Class for Word Scoring Model

In Figure 22, the word scoring model (AFINN) achieves high recall for Negative sentiment (0.9), showing strong ability to detect true negatives. Recall drops below 0.7 for Neutral and just under 0.8 for Positive sentiment, indicating that some true instances are missed. Overall, the model shows strong recall performance, particularly for Negative sentiment, with room for improvement in Neutral and Positive classes.

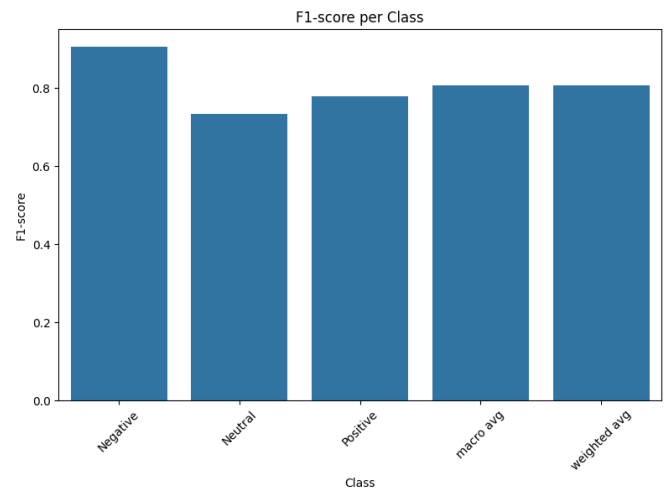


Fig. 23. F1-Score per Class for Word Scoring Model

In Figure 23, the F1-scores for the word scoring model (AFINN) show strong performance for Negative sentiment, with a score above 0.8, indicating a good balance of precision and recall. The Neutral class lags behind with an F1-score below 0.7, reflecting difficulty in handling ambiguity. Positive sentiment scores just under 0.8, suggesting solid but

improvable performance. Overall, the model handles Negative sentiment best, with room to improve for the other classes.

Both the macro average and weighted average F1-scores are close to 0.8, showing solid overall performance across all sentiment classes.

3) Precision, Recall, and F1-Score Comparison

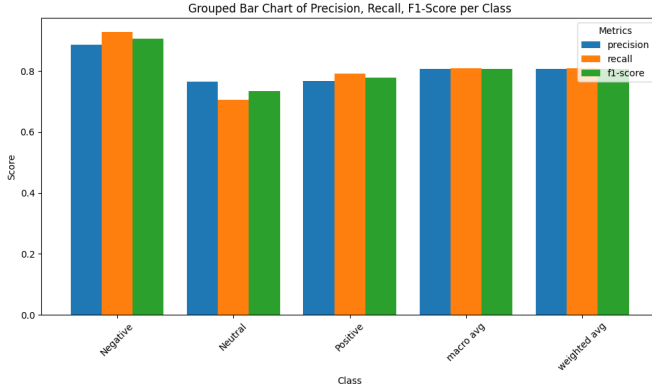


Fig. 24. Precision, Recall, and F1-Score Comparison for Word Scoring Model

In Figure 24, the grouped bar chart illustrates the word scoring model's performance across sentiment classes. It excels at detecting Negative sentiment, with precision above 0.8 and recall at 0.9. Performance drops for Neutral sentiment, where all metrics fall below 0.8, reflecting challenges with ambiguity. Positive sentiment shows more balanced results, with precision and recall near 0.8. Overall, macro and weighted averages around 0.8 indicate strong performance, though Neutral detection remains an area for improvement.

4) Accuracy

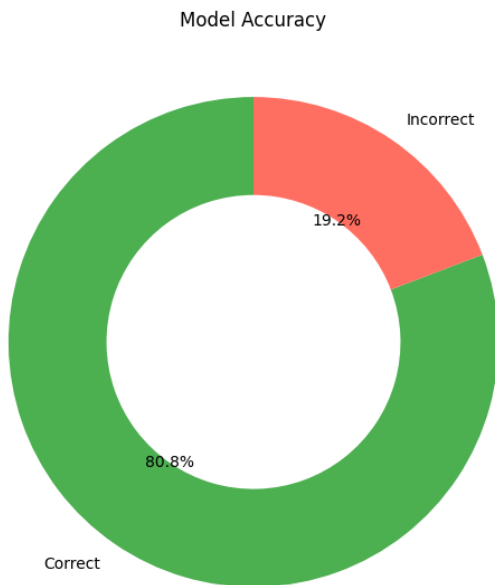


Fig. 25. Overall Accuracy of Word Scoring Sentiment Method

In Figure 25, the word scoring approach achieved a strong accuracy of 80.8%. Using AFINN to assign and aggregate token-level sentiment values, it offers a transparent and interpretable baseline. While it handles straightforward sentiments well, it may struggle with nuance and complex language. Still, its high accuracy shows that simple word-based methods remain effective for many sentiment analysis tasks.

5) Comparative Analysis

The model shows good precision for Negative sentiment (0.8253), but recall is a limiting factor (0.653). For Neutral sentiment, the model has strong recall (0.768), correctly identifying most neutral sentiments, though with some misclassification. The performance for Positive sentiment is well-balanced, with slightly better precision than recall.

In conclusion, the word scoring model demonstrated the strongest overall performance, achieving an accuracy of 80.8% and balanced precision, recall, and F1-scores across sentiment classes. It significantly outperformed the rule-based model, especially in handling neutral sentiment, though it still exhibited occasional confusion between positive and neutral categories.

D. Comparison of All Three Models

In this subsection, we compare the performance of the three sentiment analysis models: deep learning, rule-based, and word scoring. The comparison is based on key evaluation metrics, including accuracy, precision, recall, and F1-Score for each sentiment class (Negative, Neutral, Positive). Each model has its own set of advantages and trade-offs, making the evaluation crucial for understanding their suitability for different tasks.

Table I and Table II below summarize the performance of the three models.



Fig. 26. Accuracy Comparison of Sentiment Analysis Models

Figure 26 shows a comparison of the accuracy across the three sentiment analysis models: deep learning, rule-based, and word scoring. The word scoring model is the best performing, followed by deep learning, with the rule-based model coming in third.

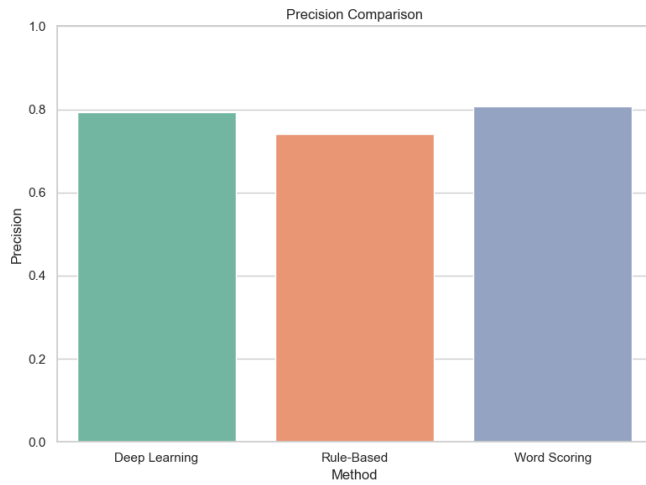


Fig. 27. Precision Comparison of Sentiment Analysis Models

Figure 27 illustrates the precision comparison across the three sentiment analysis models: deep learning, rule-based, and word scoring. The word scoring model performs the best, followed by deep learning, with the rule-based model ranking third.

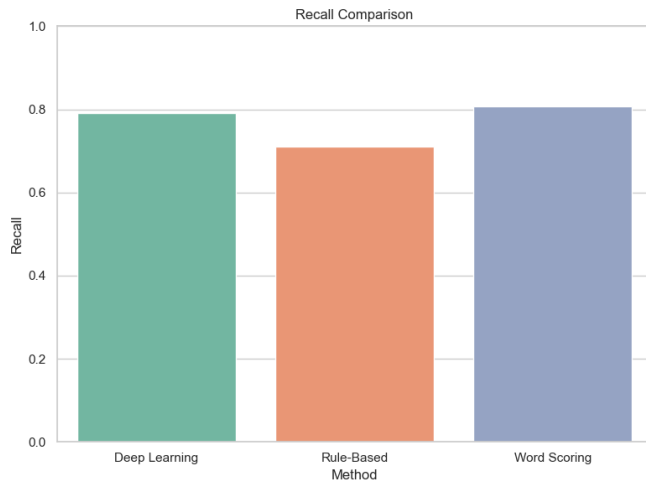


Fig. 28. Recall Comparison of Sentiment Analysis Models

Figure 28 shows the recall comparison across the three sentiment analysis models: deep learning, rule-based, and word scoring. The deep learning model outperforms the others, followed by the word scoring model, with the rule-based model performing the worst.



Fig. 29. F1-Score Comparison of Sentiment Analysis Models

Figure 29 shows the F1-score comparison across the three sentiment analysis models: deep learning, rule-based, and word scoring. Word scoring has the highest F1-score, followed by deep learning, with the rule-based model performing the lowest.

TABLE I
COMPARISON OF SENTIMENT ANALYSIS MODELS - ACCURACY AND PRECISION

Model	Accuracy	Prec Neg	Prec Neu	Prec Pos
Deep Learning	79.1%	0.91	0.63	0.76
Rule-based	71%	0.79	0.58	0.73
Word Scoring	73.6%	0.83	0.62	0.76

TABLE II
COMPARISON OF SENTIMENT ANALYSIS MODELS - RECALL

Model	Recall Neg	Recall Neu	Recall Pos
Deep Learning	0.93	0.72	0.72
Rule-based	0.69	0.80	0.64
Word Scoring	0.65	0.77	0.72

TABLE III
COMPARISON OF SENTIMENT ANALYSIS MODELS - F1-SCORE

Model	F1 Neg	F1 Neu	F1 Pos
Deep Learning	0.92	0.67	0.82
Rule-based	0.80	0.67	0.76
Word Scoring	0.82	0.70	0.74

The evaluation of the three sentiment analysis models—deep learning, rule-based, and word scoring—revealed clear distinctions in their performance, strengths, and trade-offs. Among the three, the word scoring model achieved the highest overall performance, with an accuracy of 80.8%, precision of 0.81, recall of 0.81, and F1-score of 0.81. This strong performance indicates that the model effectively balances identifying sentiment correctly while minimizing false positives and negatives. Despite being lexicon-based and not

relying on deep contextual understanding, it excels at capturing sentiment intensity and offers a practical blend of accuracy and computational efficiency.

The deep learning model, based on a transformer architecture, followed closely with an accuracy of 79.1%, precision of 0.79, recall of 0.79, and F1-score of 0.79. While slightly behind the word scoring model in raw metrics, it demonstrated more consistent performance across all sentiment classes, particularly excelling in detecting negative sentiment. Its ability to model complex linguistic patterns contributes to its strong generalization, though this comes at the cost of increased computational resources and longer processing times. Nonetheless, its robustness makes it a strong candidate for applications where subtlety and nuance in language are critical.

In contrast, the rule-based model delivered an accuracy of 71.0%, with precision of 0.74, recall of 0.71, and F1-score of 0.71. While this approach benefits from simplicity, speed, and ease of interpretation, it falls short in more ambiguous or context-dependent scenarios, particularly in correctly identifying neutral sentiment. Its reliance on static lexicons and hand-crafted rules limits its flexibility, and it tends to misclassify phrases involving sarcasm, negation, or informal language.

Overall, this comparison highlights the trade-offs between interpretability, efficiency, and accuracy. The word scoring model emerges as a strong all-around performer, offering high accuracy without the overhead of deep learning. The deep learning model, while slightly less performant in this test, offers more consistent handling of linguistic complexity and may outperform in more nuanced or domain-specific datasets. The rule-based model, while limited, remains useful for applications where transparency and speed are more critical than perfect accuracy.

These results emphasize that model selection should be driven by the specific needs of the task—whether that’s raw performance, interpretability, or operational constraints—and point to opportunities for future work to improve sentiment classification, especially in handling neutral and ambiguous content.

IX. LIMITATIONS AND DISCUSSION

When building a sentiment analysis system, there are several challenges to consider. One major issue is identifying the subjective parts of a text, where words can have different meanings depending on their context. For example, the word “crude” in “crude oil” has a different meaning than in “crude language.” Another challenge is figuring out where emotions are coming from in a sentence. Take “I was very happy to see them”—the word “they” refers to the source of the emotion, but it can be tricky to identify in more complex sentences.

Sarcasm is another tough nut to crack. Sentences like “He has gone from extreme ugliness to indescribable beauty” could easily be misinterpreted if the system doesn’t recognize the sarcastic tone. Neutral phrases or indirect negative emotions can also trip up the system. For example, the sentence “A car is better than a motor” might confuse the model, because it has to understand the subtle comparison being made.

Entity identification is another hurdle, especially with models like bag-of-words, which might fail to differentiate between entities such as “Samsung” and “iPhone.” Texts that contain errors, bad grammar, or spelling mistakes can further complicate things, as the system might misinterpret them. There’s also the challenge of working across different languages—many sentiment analysis tools are mainly built for English or Chinese, so they don’t perform as well for other languages. Finally, detecting spam comments or figuring out who originally said a quote can throw off sentiment classification and lead to inaccuracies [15].

A. Rule-based Model Limitations

The rule-based approach is simple, efficient, and easy to understand, but it does come with its drawbacks. It relies on predefined sentiment lexicons and manually crafted rules, which can’t fully capture the complexities of natural language. For example, informal language or social media posts often use sarcasm, idioms, or negation—things the model might miss. A phrase like “I just love waiting in line for two hours” could easily be misinterpreted as a positive sentiment, even though it’s meant sarcastically.

Another challenge is that these models struggle to keep up with the ever-changing nature of language. New words or phrases pop up regularly, and updating the model to handle them can be time-consuming and labor-intensive. SentiWordNet, a lexicon-based sentiment analysis tool, assigns sentiment scores (positive, negative, and objective) to synsets in WordNet, a large lexical database of English words. It tokenizes input text and retrieves sentiment scores for each word’s synset, ultimately calculating an overall sentiment score by combining these individual scores. While this works well for general sentiment analysis, it still has trouble with things like negation, sarcasm, and context-specific meanings. For instance, the phrase “The cycle is not that bad” might be incorrectly assigned a negative score, even though the intended sentiment is positive.

To improve this, methods like using n-grams (bigrams or trigrams) and part-of-speech (POS) tagging can be incorporated. These improvements help account for the context of words, making the sentiment prediction more accurate [16].

B. Word Scoring Model Limitations

The word scoring model offers an improvement over rule-based approaches by providing more tangible and interpretable results. However, it still struggles with understanding context. Essentially, this model treats text as a set of sentiment-labeled words but doesn’t account for how those words interact with each other or their arrangement within a sentence. For example, phrases like “not bad” and “bad” might receive similar sentiment scores, which is obviously incorrect. The model also has difficulty with more ambiguous situations, particularly when dealing with neutral or mixed sentiments. While it’s an improvement over basic rule-based methods, it still falls short in understanding word order or the broader context of a sentence.

This study delves into a multi-classification problem by evaluating different machine learning models using a lexicon-based text analysis approach. Lexicon-based methods, such as TextBlob and Vader Sentiment, are used to analyze sentiment in datasets, offering valuable insights into public opinions for applications like product reviews, political discussions, and more. The research underscores the influence of specific features (words) on sentiment predictions and emphasizes the importance of feature engineering in boosting model performance. Through experimental analysis, the study compares five machine learning models, with the random forest model achieving the highest accuracy of 81%. The study also applies sentiment analysis to Twitter data, using the Tweepy library to gather and analyze public opinions [17].

C. Deep Learning Model Limitations

While deep learning has made remarkable progress in areas like speech recognition and machine translation, it still faces significant challenges. One major limitation is its inability to model causal relationships, perform logical reasoning, or grasp abstract concepts. These shortcomings have led even key figures in the field to question fundamental aspects, like backpropagation, and call for the development of entirely new approaches [18]. Despite its impressive achievements, deep learning still struggles with core human-like abilities, such as causal reasoning, object understanding, and generalization. It often needs vast amounts of data and computational power just to replicate basic tasks that humans perform effortlessly.

There are promising advancements, though, including recurrent networks, self-supervised learning, graph networks, and multi-modal architectures. These innovations aim to address some of deep learning's limitations by adding memory, structure, and inductive bias. However, for true human-like intelligence to emerge, we may need to look beyond today's deep learning methods [19].

Currently, deep learning systems lack the self-organization, emergent behavior, and interpretive feedback loops that are key to human cognition. These systems depend heavily on externally curated data and fixed structures, limiting their ability to form dynamic goals or interact meaningfully with the world around them. As such, deep learning remains more of a powerful tool rather than a genuinely intelligent system. [20]

D. General Limitations Across All Models

Several systemic challenges apply across all three modeling approaches:

- **Ambiguity in Human Language:** Sentences often carry mixed or vague sentiments that cannot be easily categorized into binary or ternary classes. For instance, phrases like "It's okay, I guess" express a sentiment that is neither entirely positive nor negative, making it difficult for models to handle such nuanced or ambiguous emotions. As a result, existing models often struggle to accurately classify sentiments at the sentence level, limiting their robustness in real-world applications.

- **Neutral Sentiment Detection:** Identifying neutral sentiment is a common challenge for sentiment analysis models, especially for rule-based and deep learning methods. The neutral class is often underrepresented in training data or poorly defined, leading to a bias in the model's predictions toward more polarized sentiments. Consequently, models tend to either misclassify neutral statements or fail to identify them altogether, affecting the overall performance and fairness of sentiment classification.
- **Domain Dependence and Transferability:** Models trained on general sentiment datasets often face difficulties when applied to specific domains such as airlines, healthcare, or finance. These industries feature domain-specific language and context that general models are not well-equipped to handle. Without adaptation or techniques like domain transfer learning or few-shot learning, the scalability of these models across different sectors becomes limited.
- **Explainability and Trust:** While traditional sentiment analysis models, such as rule-based systems, offer transparency, deep learning models—particularly transformer-based architectures—are often seen as black boxes. These models make predictions without offering clear insights into the reasoning behind their decisions. This lack of interpretability is particularly concerning in domains like healthcare and finance, where understanding the model's decision-making process is crucial for building trust and ensuring accountability.
- **Computational and Environmental Cost:** Deep learning models, while powerful, demand extensive computational resources for training and fine-tuning. This results in high costs related to infrastructure and energy consumption, raising concerns about the sustainability of such approaches. The resource-intensive nature of deep learning also amplifies the environmental impact, prompting calls for more efficient and eco-friendly AI models.

E. Discussion and Implications

The comparative results underscore a clear trade-off among interpretability, performance, scalability, and deployment readiness:

- **Rule-based models** are best suited for applications requiring transparency and fast, lightweight inference—such as embedded systems or regulatory environments—but fall short in accuracy and flexibility.
- **Word scoring models** provide a middle ground in terms of explainability and performance, but their context-blind nature limits utility in complex or evolving language settings.
- **Transformer-based models** lead in predictive accuracy and generalization but introduce significant concerns around opacity, computational load, and fairness across domains or linguistic groups.

X. CONCLUSION

This study conducted a comprehensive evaluation of three sentiment analysis approaches—rule-based, word scoring, and transformer-based deep learning—using a balanced test set and standard performance metrics, including accuracy, precision, recall, and F1-score. Each method offered distinct advantages and limitations that reflect the broader trade-offs between simplicity, performance, and interpretability in sentiment classification.

The rule-based model stood out for its ease of use and transparency, making it suitable for rapid deployment and scenarios requiring interpretability. However, its heavy reliance on predefined lexicons and static rules made it less effective at handling linguistic ambiguity, especially with neutral sentiment. As a result, it achieved the lowest accuracy of 71%, reflecting limitations in flexibility and contextual understanding.

In contrast, the word scoring model demonstrated the strongest overall performance, achieving an accuracy of 80.8%, with balanced and high precision, recall, and F1-scores. Despite being lexicon-based, it was more capable of quantifying sentiment intensity and mitigating some of the rule-based model's limitations. This model effectively bridged the gap between interpretability and performance, offering a practical solution with low computational demands.

The transformer-based deep learning model, leveraging the cardiffnlp/twitter-xlm-roberta-base-sentiment architecture, achieved an accuracy of 79.1%. It outperformed the rule-based method and performed comparably to the word scoring model, particularly excelling in handling negative sentiment. Its strength lies in its ability to model complex language patterns and generalize well across classes. However, it was computationally more demanding and showed slightly weaker reliability when classifying neutral sentiments—an area that remains challenging across all models.

Overall, the results underscore that the optimal sentiment analysis approach depends on the specific use case. While deep learning offers superior modeling capabilities, traditional methods retain value when speed, transparency, and lower resource use are prioritized. Importantly, the study highlights a consistent gap in accurately identifying neutral sentiment, a limitation that warrants further research and innovation.

In conclusion, this study provides meaningful insights into the comparative performance of different sentiment analysis techniques. It emphasizes the importance of balancing accuracy with interpretability and resource efficiency and paves the way for developing more adaptable and explainable sentiment analysis systems that can better meet the diverse needs of real-world applications.

XI. FUTURE WORK

Future work can explore several directions to address current limitations and enhance sentiment analysis systems:

- **Neutral Sentiment Handling:** Develop targeted classifiers or leverage contrastive learning to better capture the

subtle cues of neutral expressions. This could involve incorporating syntactic ambiguity detection or sentiment trajectory analysis over time.

- **Hybrid Models:** Explore ensemble approaches that combine rule-based or statistical heuristics with transformer embeddings to blend interpretability and predictive power. Dynamic weighting mechanisms or attention-based integration could further refine these models' adaptability to various domains.
- **Domain Adaptation:** Fine-tune sentiment models on airline and airport-specific data or similar customer review domains to improve contextual accuracy. This includes leveraging weak supervision or self-training to utilize unlabeled domain data effectively.
- **Explainability in Deep Learning:** Apply explainable AI techniques (e.g., SHAP, LIME, attention visualization) to illuminate the internal decision-making of complex models and build trust in automated sentiment systems. Future research could investigate interactive explanation interfaces or user-in-the-loop correction mechanisms.
- **Multilingual and Cross-Cultural Sentiment Analysis:** Broaden the scope by including sentiment data from different languages and cultural backgrounds, thereby improving inclusivity and generalization. This may also require the development of culture-sensitive sentiment lexicons or regional fine-tuning strategies.
- **Real-Time and Scalable Deployment:** Investigate model compression, quantization, or knowledge distillation strategies to enable faster inference and deployment in production environments. Edge-optimized transformer variants and streaming data architectures could be explored for low-latency scenarios.
- **Temporal Sentiment Dynamics:** Analyze how sentiment evolves over time in user behavior, particularly for longitudinal customer engagement or brand perception tracking. This may involve sequential models or time-aware embeddings.
- **Multi-Modal Sentiment Analysis:** Integrate textual sentiment analysis with other modalities such as images, audio, or video to better capture affective cues in social media, product reviews, or customer service interactions.
- **Adversarial Robustness and Bias Mitigation:** Study how sentiment models behave under adversarial inputs or biased data distributions. Future work could focus on training techniques to improve model resilience and fairness across demographic groups.
- **Human-AI Collaboration:** Design sentiment analysis tools that augment rather than replace human judgment—supporting decision-making in areas like customer support, market research, or mental health analysis. Interactive dashboards or semi-automated labeling pipelines could be valuable here.

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APPENDIX

Python Environment and Libraries

The following libraries and their respective versions were installed in the Python environment used for all experiments:

- `afinn==0.1`
- `aiohappyeyeballs==2.6.1`
- `aiohttp==3.11.17`
- `aiosignal==1.3.2`
- `attrs==25.3.0`
- `beautifulsoup4==4.13.4`
- `certifi==2025.1.31`
- `charset-normalizer==3.4.1`
- `click==8.1.8`
- `colorama==0.4.6`
- `contourpy==1.3.2`
- `cycler==0.12.1`
- `filelock==3.18.0`
- `fonttools==4.57.0`
- `frozenset==1.6.0`
- `fsspec==2025.3.2`
- `huggingface-hub==0.30.2`
- `idna==3.10`
- `Jinja2==3.1.6`
- `joblib==1.4.2`
- `kiwisolver==1.4.8`
- `MarkupSafe==3.0.2`
- `matplotlib==3.10.1`
- `mpmath==1.3.0`
- `multidict==6.4.3`
- `networkx==3.4.2`
- `nlTK==3.9.1`
- `numpy==2.2.5`
- `openai==0.28.0`
- `packaging==25.0`
- `pandas==2.2.3`
- `pillow==11.2.1`
- `propcache==0.3.1`
- `pyparsing==3.2.3`
- `python-dateutil==2.9.0.post0`
- `python-dotenv==1.1.0`
- `pytz==2025.2`
- `PyYAML==6.0.2`
- `regex==2024.11.6`
- `requests==2.32.3`
- `safetensors==0.5.3`
- `scikit-learn==1.6.1`
- `scipy==1.15.2`
- `seaborn==0.13.2`
- `setuptools==80.1.0`
- `six==1.17.0`
- `soupsieve==2.6`
- `sympy==1.13.1`
- `textblob==0.19.0`
- `threadpoolctl==3.6.0`
- `tokenizers==0.21.1`

- `torch==2.6.0`
- `tqdm==4.67.1`
- `transformers==4.51.2`
- `typing_extensions==4.13.2`
- `tzdata==2025.2`
- `urllib3==2.4.0`
- `vaderSentiment==3.3.2`
- `yarl==1.20.0`

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